

Reverse Neighbor Sliding and Order Selection for Efficient Multi-Proximity Graph Merging

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Abstract

Approximate k Nearest Neighbor (AKNN) search in high-dimensional space is a foundational problem in vector databases with widespread applications. Among the numerous AKNN indexes, Proximity Graph-based indexes achieve state-of-the-art search efficiency across various benchmarks. In many real-world scenarios, datasets are maintained as multiple segment-level graph indexes to support continuous writes and segment management. However, these fragmented indexes complicate maintenance and degrade search efficiency, making fast graph index merging essential. In this paper, we focus on the efficient merging of multiple existing graph indexes into a single one. To achieve this, we propose a Reverse Neighbor Sliding Merge (RNSM) that exploits structural information to boost merging efficiency. We further propose Merge Order Selection (MOS) to minimize total merge cost across multiple indexes by eliminating redundant operations. Experiments show that our approach yields up to a $3.86\times$ speedup over existing index merge methods and a $9.92\times$ speedup over index reconstruction, while maintaining superior search performance. Moreover, our method scales to merging up to 50 sub-indexes on datasets of 100 million vectors, maintaining consistent speedups.

CCS Concepts

• Information systems → Information retrieval.

Keywords

Approximate Nearest Neighbor Search, Graph Index, Vector Databases, Information Retrieval

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1 Introduction

The k nearest neighbor (KNN) search in high-dimensional space is a fundamental problem that has received widespread attention due to its extensive applications in data mining [9], information retrieval [37, 45], recommendation systems [10, 47], and the Retrieval-Augmented Generation (RAG) of Large Language Models (LLMs) [34].

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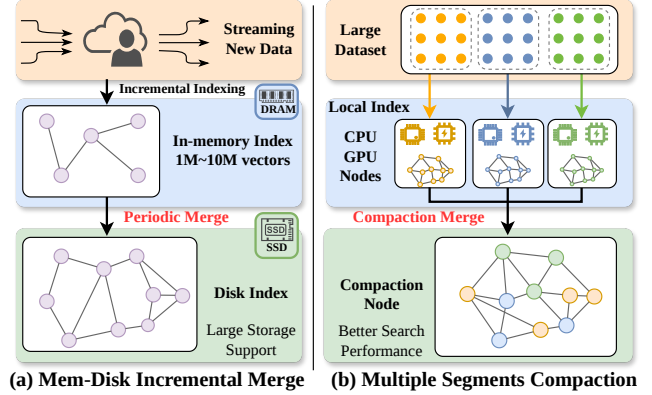


Figure 1: Examples of Multiple Indexes Merging Scenarios

However, the cost of exact KNN search is extremely high for large-scale datasets [27]. Therefore, extensive research has focused on studying approximate k nearest neighbor (AKNN) search [1–3, 5, 7, 8, 11, 12, 14–23, 25, 30, 35, 38, 39, 42, 44, 46, 48, 52–55, 63, 67, 68], trading accuracy for search efficiency. Among various AKNN methods, graph-based indexes are the state-of-the-art solution, demonstrating superior performance compared to other approaches [4, 36, 41, 60]. Graph-based methods represent each vector in the dataset as a node in a graph, where edges connecting nodes reflect proximity relationships designed to facilitate efficient navigation during search. Most graph-based methods construct a Proximity Graph (PG) through one of two approaches: (1) refining a random or approximate k nearest neighbor graph (KNNG) [15, 16, 44, 70], or (2) incrementally inserting nodes into an empty graph [38, 39].

However, both indexing methods incur substantial time costs for graph construction [4, 60] due to extensive distance computations. Therefore, commercial vector databases like Milvus [58] and BlendHouse [43] typically partition data *by design* to support dynamic and independent segment data management. Despite the scalability of this approach, directly searching multiple sub-indexes incurs a significant search performance degradation. This phenomenon is intuitive because searching m separate graph indexes independently requires traversing each from its entry point. Given that graph search complexity is logarithmic with respect to dataset size [16, 35, 44], querying m partitions scales the computational cost by a factor of approximately m . As shown in the experiment results in Fig. 2, the search efficiency of the partitioned index with 10 partitions is only half that of the complete index, and this efficiency gap widens to nearly 1/6 with 50 partitions. Therefore, merging multiple indexes is a critical operation in various scenarios.

Fig. 1 demonstrates two scenarios that rely on index merging. The first one is the *incremental index merging* when the new data keeps coming in streams. As shown in Fig. 1(a), to handle continuous data writes while managing the memory-intensive storage overhead of graph indexes, commercial systems typically adopt

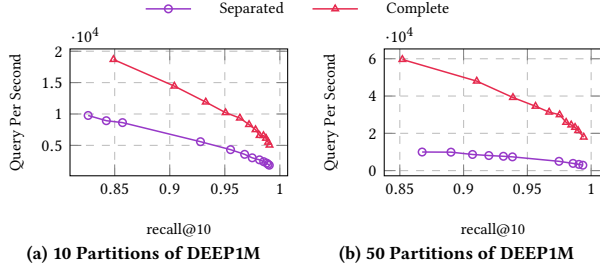


Figure 2: Separated index search performance. Sub-indexes are constructed on 10 and 50 equal-sized partitions of the DEEP1M dataset, and searching is conducted sequentially on all partitions with result merging. It shows that (i) the separated index has a significant performance gap compared with the complete index, and (ii) this gap widens as the partition number increases.

a memory-disk hybrid architecture: large-scale graph indexes reside on SSD, while newly arriving data are incrementally indexed in memory (typically fewer than 10M vectors). Periodically, these in-memory indexes are offloaded and merged into the disk-based indexes guided by the Log-Structured Merge (LSM) architecture [43]. However, existing methods [50, 79] handle merging in the same way as incremental indexing, which is computationally expensive. The second one is the *multiple segment compaction* when a large index is constructed in distribution. As shown in Fig. 1(b), it utilizes distributed storage and indexing systems for better indexing efficiency [71, 75]. In these deployments, large datasets are partitioned into multiple segments, with each CPU or GPU node independently building and searching its local index. However, periodic compaction becomes necessary to reduce storage overhead, simplify management, and improve search performance. Existing systems [49, 75] typically handle compaction with a full rebuild from scratch, which introduces high computational cost. Therefore, speeding up index merging is crucial for index performance.

Previous Methods. We categorize existing index merge methods into two families based on their strategies for connecting multiple sub-graph indexes: (1) **Overlap-based** [52, 66, 72] methods intentionally duplicate data points across multiple segments during sub-index construction. The sub-indexes can be merged by concatenating the neighbor lists of the overlapping nodes. (2) **Search-based** [39] treat the source index data as queries to search in the target index. The neighbors of the subgraph index node are updated with the search results, which can be viewed as an incremental insert. Recent works [32, 64, 65, 76] focus on accelerating search-based methods, as they are distribution-agnostic and directly benefit from ANNS query optimizations. These methods focus on reducing the time cost of merging multiple existing graph indexes in main memory by reducing the Number of Distance Computations (NDC). This is a practical setting with growing DRAM capacity [59, 60] and distance computation dominating the cost.

State-of-the-art. A recent search-based study, FGIM [64], has emerged as a state-of-the-art method for efficient index merging. FGIM treats the index merging process as a KNNG refinement problem, *i.e.*, discovering the KNNs for every vector within a given dataset. Specifically, FGIM first performs PGs to KNNG transformation based on cross-querying: given a set of PGs, each node is

searched for ANNs in all other PGs, and the search results are used to construct a KNNG. To mitigate the high computational cost of cross-querying, FGIM proposed a Minimum Candidate Set strategy that bounds the minimum required cross-graph candidates based on the assumption that each partition contributes independently and equally. Then, KNNG refinement improves the initial candidate neighbor sets by iteratively identifying high-quality neighbors. Finally, the refined KNNG is transformed back to a PG by applying the RNG-based neighbor selection rule. However, this Minimum Candidate Set mechanism of FGIM faces significant challenges: (1) When merging only two indexes, the strategy degenerates to exhaustive cross-querying without candidate reduction benefits; (2) The theoretical assumption limits its applicability to imbalanced partitions, where few sub-indexes could contribute significantly more cross-neighbors than others; and (3) The strategy requires the candidates set size to exceed the number of partitions, which increases the computational cost of KNNG refinement when handling a large number of partitions.

Our Solution. In this paper, we propose a novel search-based multi-index merging framework that addresses the above limitations through two key techniques: *Reverse Neighbor Sliding Merge* (RNSM) and *Merge Order Selection* (MOS). We first focus on efficiently merging two graph indexes and develop RNSM, a parallel index merging method with key innovations by leveraging previous search results to accelerate query processing. RNSM is built on three key components: *Neighbor Expand*, *Pivot Selection*, and *Sliding Merge*. RNSM explicitly exploits the structural information of both PG indexes via neighbor expansion. We then greedily select a set of *pivots* from the source index. By organizing the pivots and their neighbor, RNSM performs parallel-friendly local sliding via the reuse process of pivot search results, substantially improving merge efficiency. On top of RNSM, we introduce MOS as a practical optimization for multi-index merge order. We base MOS on two key observations: (1) naive pairwise merging introduces redundant operations, and (2) the connectivity pattern between graph indexes strongly influences the search performance of the final merged index. With the above key observations, we design the MOS strategy to compute a sparse merge order graphs that (i) reduce unnecessary merge operations and (ii) preserve or even enhance query efficiency in the final merged index.

Contribution. We summarize our main contributions as follows:

Problem Analysis and Motivation (§ 3). We explored the key factors in index merging and found that nearest neighbor connections are crucial, thus transforming the index merging problem into a search optimization problem. We also analyzed the patterns of multi-index merging, providing new insights into the merge order selection.

Efficient Two-Index Merge Method (§ 4). We propose RNSM, an efficient two-index merge method that leverages structural information from both the source and target indexes to accelerate merging. RNSM identifies pivots in the source index by constructing approximate reverse nearest neighbor relationships. The search results of each pivot are then propagated to its followers to perform efficient local sliding.

Table 1: A Summary of Notations

Notation	Description
X_i	A set of points/vectors
x, y	A data point/vector
$\mathbb{R}^D$	D -dimensional Euclidean space
$\delta(x, y)$	Euclidean/ L_2 distance between x and y
$G_i = (V_i, E_i)$	The graph index for X_i with vertex set V_i and edge set E_i
$\mathcal{L}(p, q)$	The local sliding cost from point p to q
$N(v, G)$	The neighbors of v in G
$N_k(x, X)$	The k -nearest neighbors of x in X
$R_k(x, X)$	The reverse k -nearest neighbors of x in X
R	The maximum out-neighbors of any node in the graph index

Merge Order Selection (§ 5). We propose MOS, which formulates merge order selection as a graph optimization problem with dual objectives: minimizing merge cost through sparsity maximization and preserving search quality through diameter control. We introduce tailored strategies for both random and cluster-based partitions for different application scenarios.

Extensive Experiment Analysis (§ 6). We conduct comprehensive experiments on real-world datasets for multi-index merging. We integrate our framework into popular graph-based indexes, including HNSW, NSG, SSG, and τ -MNG, demonstrating its broad applicability. Our method delivers up to 9.92 $\times$ speedup while preserving search quality and efficiency. Moreover, the speedup ratio remains stable as the number of partitions increases with 50 sub-indexes and 100 million data, indicating robust scalability.

Due to space constraints, some additional proofs and experiments are omitted, which can be found in the Appendix.

2 Preliminaries

We first formally define the AKNN search problem in Section 2.1 and then explain the graph-based AKNN index in Section 2.2.

2.1 The AKNN Search Problem

Table 1 lists the frequently used notations and their definitions. We first describe the KNN search problem:

Definition 2.1 (KNN Search). Given a dataset X with n vectors in the D -dimensional Euclidean space $\mathbb{R}^D$, and a query vector $q \in \mathbb{R}^D$. Let $\delta(x, y)$ denote the distance between two D -dimensional vectors x and y . The KNN search retrieves k vectors $\text{KNN}(q)$ in X such that $\delta(x, q) \leq \delta(x', q), \forall x \in \text{KNN}(q)$ and $x' \notin \text{KNN}(q)$.

With the rapid growth of dataset dimensionality and scale, exact KNN search fails to meet the efficiency requirements, known as the curse of dimensionality [27]. Therefore, most efforts have focused on the approximate AKNN search, which trades a small loss in accuracy for a significant gain in efficiency. Recall@K is commonly used as a metric to evaluate the precision of the AKNN search algorithm. Recall@K is defined as the ratio of the true k nearest neighbors that appear in the k nearest results returned by the AKNN search algorithm, and it is computed as follows:

$$\text{Recall@K} = \frac{|\text{AKNN}(q) \cap \text{KNN}(q)|}{K} \quad (1)$$

where $\text{AKNN}(q)$ is the top- K nearest results returned by the AKNN search, and $\text{KNN}(q)$ is the actual KNN for a given q . Based on

Algorithm 1: AKNN Beam Search

Input: Graph Index $G = (V, E)$, Entry Points $\{ep\}$, Query q , Candidate Set Size ef , Result Number k
Output: Top- k approximate nearest neighbors of q

- 1 Initialize priority queue $C \leftarrow \emptyset$;
- 2 $C \leftarrow \{(ep, \delta(ep, q))\}$;
- 3 Each node $v \in V$ set $v.\text{expanded} \leftarrow \text{false}$;
- 4 **while** C has unexpanded nodes **do**
- 5 $u \leftarrow$ nearest unexpanded node in C ; $u.\text{expanded} \leftarrow \text{true}$;
- 6 **for each** unexpanded neighbor v of u **do**
- 7 $C \leftarrow C \cup \{(u, \delta(u, v, q))\}$;
- 8 **if** $|C| > ef$ **then**
- 9 resize C to size ef ;
- 10 **return** k nearest nodes in C ;

the above definition, numerous AKNN Search Indexes have been proposed with the aim of achieving higher recall at a lower cost.

2.2 Graph-Based AKNN Search Index

Most AKNN search algorithms find the approximate nearest neighbors via well-designed indexes, which can be divided into four categories: hash-based [12, 17, 17, 22, 48, 53–55, 62, 63], quantization-based [1, 2, 5, 18–21, 30, 67, 68], tree-based [7, 11, 14, 42], and graph-based [3, 8, 15, 16, 23, 25, 31, 38, 39, 44, 51, 52, 57, 59, 78]. Among them, the Graph-based methods have gained prominence due to their state-of-the-art search efficiency. Specifically, the Graph-based AKNN indexes take each vector in X as a node on the graph and construct a Proximity Graph (PG) to represent the similarity relationships between vectors. Then, given a dataset X with n vectors in the D -dimensional Euclidean space $\mathbb{R}^D$, we construct a graph index $G = (V, E)$ for X with vertex set V and edge set E : Every $v \in V$ corresponds to a unique vector x in X , and an edge $(u, v) \in E$ denotes the neighbor relationship between $u, v \in V$.

Most graph-based methods utilize a beam search algorithm presented in Algorithm 1 to retrieve AKNNs. During the beam search, set C is maintained as the candidate nearest neighbors of q . C is the ef nearest neighbors retrieved so far (line 1). The beam search starts from the given entry points (line 2). At each step of the search, it checks the neighborhood of the nearest unexpanded node and tries to insert these neighbors into C (lines 4–9). The algorithm terminates when no unexpanded node can be found in C (line 10).

The PG index is constructed in two stages: (1) Perform beam search in Algorithm 1 on the current index and obtain neighbor candidates W from top- $ef_{\text{construction}}$ ($ef_{\text{construction}} \gg ef_{\text{search}}$) results [38, 39] or visited nodes during searching [16, 44, 52]. (2) Apply the neighbor pruning rule of RNG (Relative Neighbourhood Graph) [42, 56] or its variants [15, 35, 44, 46, 52, 70] to W to select neighbors. Reverse edges are further added for better connectivity.

3 PROBLEM ANALYSIS AND MOTIVATION

In this section, we formalize the multi-index merging problem and present key observations that motivate our method (Section 3.1). Based on these insights, we introduce the pipeline of our merging framework (Section 3.2).

3.1 Problem Analysis and Motivation

We begin our analysis by defining the m -graph index merge problem. Given m partitions $\{X_1, X_2, \dots, X_m\}$ of dataset X such that

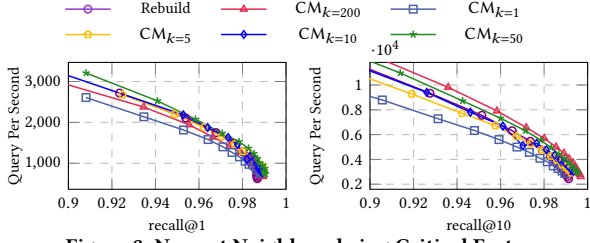


Figure 3: Nearest Neighbors being Critical Factor

$X = \bigcup_{i=1}^m X_i$ and $X_i \cap X_j = \emptyset$ for all $1 \leq i < j \leq m$, the m -graph index merge problem is defined as follows:

Problem Definition. Given m well-constructed graph indexes $I = \{G_1, G_2, \dots, G_m\}$, where $G_i = (V_i, E_i)$ and $V_i = X_i$, the m -graph index merge problem aims to construct a unified graph index $G = (V, E)$ that satisfies the following constraints:

- **Search Quality:** The merged graph G achieves comparable search performance to an index built from scratch on X .
- **Computational Efficiency:** The merging of multiple indexes requires less time than reconstructing an index for X from scratch.

The current index merging paradigm **Cross-query Merge (CM)** is based on pairwise searching: For each point x in a source index X_i , it queries all $m - 1$ target indexes $I \setminus \{G_i\}$ to retrieve a candidate neighbor set W of size ef_c , which corresponds to the construction parameter. However, since it is unknown which indexes contain the top- ef_c candidates, all $m - 1$ target indexes must be searched, incurring $m - 1$ times of ef_c -NN searches and $(m - 1) \cdot ef_c$ candidates in total. As we have $|V|$ points to merge, then it requires $O(|V|(m - 1))$ searches with $O(|V| \cdot (m - 1) \cdot ef_c)$ intermediate candidates, which is both time-consuming, and could even take longer time than reconstruction. Obviously, reducing $m - 1$ and ef_c could reduce this large complexity. In the following, we provide two observations on how we could reduce them without sacrificing index quality.

Observation 1: Nearest Neighbors are Key Contributing Components of Merging. As mentioned above, most graph-based methods construct proximity graphs using a two-stage framework, where higher-quality pruning candidates W lead to better index quality. These pruning candidates W aim to cover a large region around a given vertex, which directly leads to a high time cost. W can be divided into two parts: (1) Nearest neighbors that provide good connectivity among KNNs. (2) Distant neighbors that enable efficient graph traversal by introducing long-range edges and covering broader regions of the data space [72]. We find that **nearest neighbors are critical for merging quality**, as the distant neighbors are already captured within sub-indexes. This phenomenon can be verified in the following example:

EXAMPLE 1. We first construct an NSG index directly on the complete SIFT1M dataset with $ef_c = 200$ and denote it as **Rebuild**. Then we relax the per-target-index search budget of CM from the fixed ef_c to $k = [1, 5, 10, 50, 200]$, where 200 corresponds to the default setting. As illustrated in Fig. 3, we investigate the impact of k from target indexes on merged index quality. Specifically, incorporating nearest neighbors from other indexes dramatically improves performance even with just $k = 1$, and it achieves the identical performance of

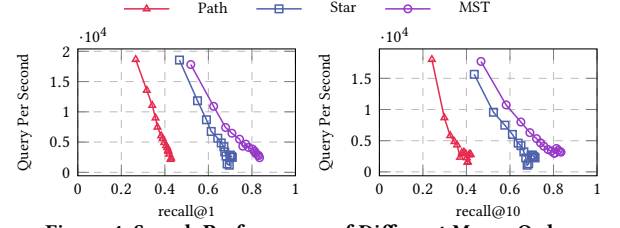


Figure 4: Search Performance of Different Merge Order

Rebuild at high recall (≥ 0.99). With larger k from target indexes, the merged index achieves a better efficiency-accuracy tradeoff.

Based on this observation, we can safely reduce the merging problem to a more efficient formulation. Standard index construction typically requires large pruning candidate sets with $|W| \geq 200$. In contrast, for merging purposes, we only need to identify the top- k nearest neighbors from the target index for each vertex. Here, k can be substantially smaller, e.g., $k \in [1, 50]$, which saves the computation significantly by limiting the search to a small number of nearest neighbors from other indexes, making the merging process substantially more efficient with smaller k .

Observation 2: Different Merge Orders Lead to Varying Search Performances. The merge order specifies which pairs of indexes to merge next. According to our analysis, retrieving KNNs from other indexes for each vertex is critical, which naturally leads to a pairwise processing framework. However, this framework suffers from inherent computational redundancy for the following reasons: (1) KNNs from different partitions may occlude one another during merging. Empirically, only 9.2% of the 100-NNs remain as neighbors of the vertex after neighbor selection. (2) For cluster partitions, KNNs of a vertex are often concentrated in a subset of partitions rather than uniformly distributed across all partitions. Moreover, we observe that **even with the same number of merge operations, different merge orders result in significantly different search performance**, as shown in the following example:

EXAMPLE 2. We consider three different merge orders: **Path** constructs a linear merging chain, **Star** merges all indexes with a central one, and **MST** builds a minimum spanning tree based on inter-index similarity. As shown in Fig. 4, despite using an identical number of merge operations ($m - 1$ for m indexes), different merge orders yield dramatically different performance. **Path** fails to reach higher recall because KNNs of a vertex may be separated across partitions at opposite ends of the linear merge sequence. **Star** and **MST** significantly improve search performance through enhanced connectivity, enabling more comprehensive cross-partition neighbor discovery. However, **Star** imposes an imbalanced merge workload on the central hub index, resulting in inferior performance compared to **MST**.

3.2 Multi-index Merging Framework Overview

Fig. 5 illustrates the pipeline of our proposed RNSM + MOS framework, which has the following three stages:

a) Sub-Indexes Preparation. While existing methods are tailored to either cluster partitions [52, 72] or random partitions [64, 66], our framework can be applied to both partition types without compromising performance. Given pre-built sub-indexes, we first prepare

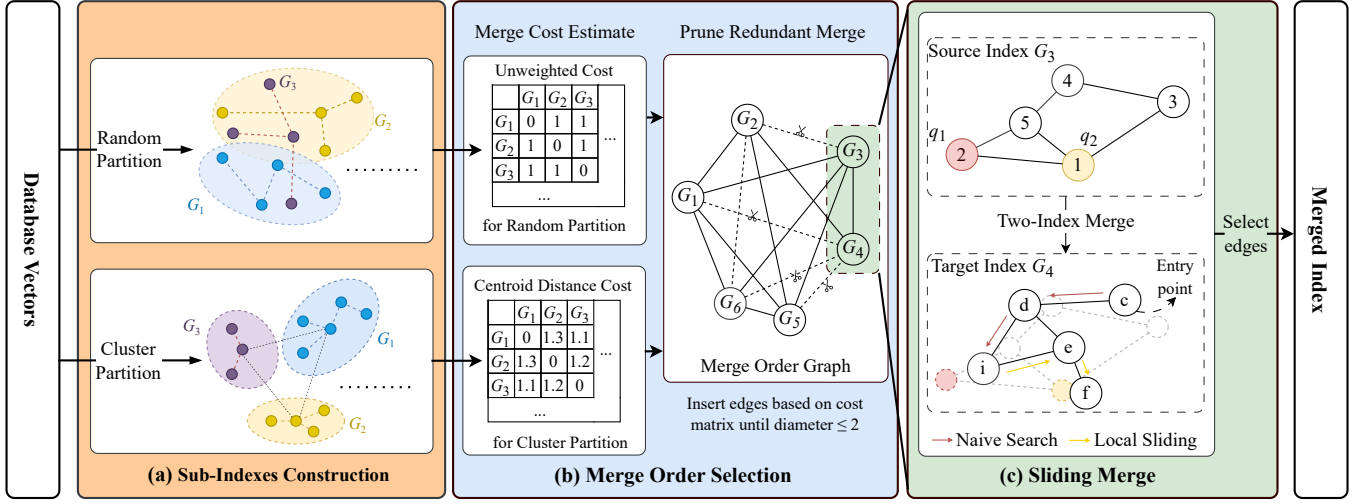


Figure 5: Pipeline of Our Multi-Index Merge Framework

the merge cost matrix based on their partition centroids. For random partitions, the merge costs for different index pairs are considered identical. For cluster partitions, we estimate the merge cost for two indexes using the distance between their centroids, as nearby indexes are more likely to share KNNs and thus benefit more from merging. Then the cost matrix is utilized for merge order selection.

b) Merge Order Selection (MOS). Next, we reduce the cost of multi-index merging through MOS. It acts as a lightweight merge planner on top of pairwise RNSM. We model merge operations as a merge order graph and formulate the problem as minimizing the number of edges for efficient merging while maintaining sufficient connectivity for high-quality search. Specifically, we use the average degree of the merge order graph to measure merge cost and the graph diameter to measure connectivity. Guided by the cost matrix, we construct a sparse merge order graph that effectively prunes unnecessary merges (e.g., G_2 and G_3) from pairwise merging.

c) Sliding Merge. Finally, we merge the adjacent indexes (e.g., G_3 and G_4) in the merge order graph using our RNSM algorithm. RNSM exploits prior search results to accelerate later searches. Specifically, we greedily select nodes with the maximum number of Reverse Nearest Neighbors (RNN) in the source index as pivots. Then, the search result i of pivot 2 in the target index is used as an entry point to find cross-index neighbors for RNNs of the pivot 2.

4 RNSM for Pairwise Indexes Merge

This section details the proposed Reverse Neighbor Sliding Merge framework for merging two graph indexes. We first formally define the concept of *sliding merge* and derive a corresponding cost model in Section 4.1 to quantify the merging overhead. Then, we introduce the RNSM algorithm in Section 4.2, demonstrating how it optimizes the merging process by minimizing the derived cost objective.

4.1 Sliding Merge Cost Model

Our preliminary analysis identifies the AKNN search within the target index as the primary bottleneck in index merging (see Fig. 3). Consequently, optimizing the search process on the target index

is paramount. Prior research established that initializing graph-based searches with entry points topologically closer to the query target significantly reduces the search path length, thereby reducing latency [16, 52, 77]. Building on this insight, we propose *Sliding Merge*, a strategy that leverages search results from neighboring nodes to accelerate processing. In this section, we formalize the cost model and define the optimization problem.

Notation and Sliding Semantics. Given a source graph index $G_1 = (V_1, E_1)$ and a target index $G_2 = (V_2, E_2)$, our objective is to merge G_1 into G_2 by performing an AKNN search in G_2 for every vector $x \in V_1$. The *Sliding Merge* framework employs two distinct search operations:

- (1) **Naive Search:** A standard AKNN search for query $x \in V_1$ starting from the default entry point of G_2 .
- (2) **Local Sliding:** A search for query x initialized using the result of a previous search for a pivot node p .

Let $\mathcal{L}(p, x)$ denote the search cost for query x . If a pivot p is used, the cost depends on the proximity between p and x ; otherwise, a fixed base cost is incurred.

Cost Metric. To ensure our model is hardware-agnostic, we employ the Number of Distance Computations (NDC) as the cost metric, a standard practice in nearest neighbor search analysis [19, 39, 52, 74, 79]. We define the cost of a *Naive Search* as a constant γ , derived from the average NDC of a search with a default entry point. For *Local Sliding*, recent findings [65] indicate that when a query x starts from a pivot p proximate in the high-dimensional space, the NDC required to converge follows a linear relationship with their distance $\delta(p, x)$ (see Fig. 6). Accordingly, we approximate the sliding cost as $\mathcal{L}(p, x) \approx \delta(p, x)$.

Problem Formulation. While local sliding offers lower costs than naive search ($\delta(p, x) < \gamma$ for nearby pivots), it necessitates a prerequisite: the pivot p must have already been searched. Ideally, every node would slide from its nearest resolved neighbor. However, identifying the optimal global pivot for x would itself require a costly search. To make this tractable, we exploit the topology of the source

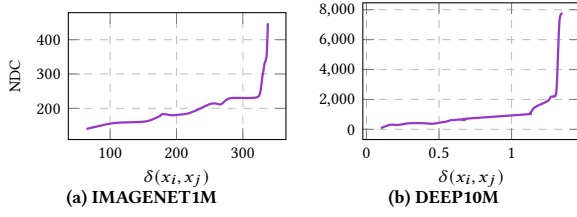


Figure 6: Verification of Distance Cost Model. In datasets IMAGENET1M and DEEP10M, the cost of sliding from x_i to x_j is linear to the distance when x_i and x_j are close [65].

graph G_1 . Since G_1 is a proximity graph, neighbors in G_1 are inherently close in the vector space [16]. Thus, we restrict the candidate pivots for a node v to its immediate neighbors $N(v)$ in G_1 . This transforms the merge optimization into a pivot selection problem: we must select a subset of nodes to perform Naive Search (pivots), such that all other nodes can perform low-cost Local Sliding from a pivot. Moreover, we require pivots to be independent of each other to fully decouple them. We formally define this as follows:

Definition 4.1 (Minimum Cost Independent Dominating Pivot Selection (IDPS)).

- **Input:** A vector dataset X , a graph $G = (V, E)$ where node $v_i \in V$ corresponds to $x_i \in X$, a distance function $\delta(\cdot, \cdot)$, and a naive search cost γ .
- **Output:** A pivot subset $P \subseteq V$ that minimizes the total cost $\mathcal{L}(P)$, subject to the independent dominating set constraint.
- **Constraint:** Every node $v \in V$ must be covered:

$$v \in P \vee \exists p \in P \text{ s.t. } (v, p) \in E$$

Additionally, P is an *independent set*: no two pivots are adjacent in G .

- **Objective:** Minimize $\mathcal{L}(P) = \mathcal{L}_{naive}(P) + \mathcal{L}_{slide}(P)$, where:

$$\mathcal{L}_{naive}(P) = \gamma \cdot |P|$$

$$\mathcal{L}_{slide}(P) = \sum_{v \in V \setminus P} \min_{p \in P \cap N(v, G)} \delta(x_v, x_p)$$

THEOREM 4.2. *The Minimum Cost Independent Dominating Pivot Selection (IDPS) problem is NP-hard.*

PROOF SKETCH. We reduce from the NP-complete Minimum Independent Dominating Set (MIDS) problem [29]. Given a MIDS instance $\langle G', k \rangle$, set $G = G'$, define the discrete metric $\delta \equiv 1$ for distinct nodes, and set $\gamma = 2$, $C = k + |V|$. For any feasible P (an independent dominating set), the cost simplifies to $\mathcal{L}(P) = 2|P| + (|V| - |P|) = |P| + |V|$. Hence $\mathcal{L}(P) \leq C$ iff $|P| \leq k$, establishing a one-to-one correspondence between MIDS solutions and IDPS solutions. The full proof is provided in the Appendix. $\square$

Given the intractability of the IDPS problem, we resort to a greedy algorithm for pivot selection presented in Algorithm 2.

4.2 Reverse Neighbor Sliding Merge

A straightforward algorithm for IDPS is to adopt a greedy strategy: iteratively selecting the pivot with the highest *unit value* (coverage nodes divided by cost). However, applying this directly to index

merging is problematic for two reasons. 1) *Limited Visibility*: Relying solely on the sparse connections in the source PG limits the ability to find the true nearest pivot, potentially forcing sub-optimal sliding paths. 2) *Computational Expensive*: Since pivots and their sliding nodes are not necessarily adjacent in the source index, calculating unit values requires either exhaustive all-pairs distance computation, or reliance on graph indexes that often store neighbor rankings rather than explicit distance values [65], compromising both flexibility and precision.

The RNSM Strategy. To overcome these limitations, we propose *Reverse Neighbor Sliding Merge (RNSM)*. The core intuition is to resolve pivot selection *globally* before merging, enabling full parallelism. Instead of dependent selection, we exploit the proximity properties of the Reverse k -Nearest Neighbor (RKNN) graph. We find that if a node x appears in the KNN list of y , then x is inherently a good pivot for y due to proximity. Consequently, nodes with a high-quality RNN graph are natural hubs that can serve as cost-efficient pivots for many neighbors. We first introduce two core components of the RNSM: Pivot Selection and Local Sliding.

Pivot Selection. We first identify the pivots and their corresponding followers to perform local sliding before executing merge. Each node in the source index G_1 is classified into:

- (1) **Pivots p :** Neighbor candidates obtained by a naive search in target index G_2 .
- (2) **Follower F_p :** Neighbor candidates obtained by local sliding from search results of pivot p .
- (3) **Uncovered U :** The remaining nodes.

As shown in Fig. 6, for the search results of one node to serve as effective entry points for another node, the two nodes must lie in proximity in the metric space. This motivates us to select nodes that are close to many other nodes as pivots. We observe that RKNNs naturally captures this notion: unlike KNNs, the exact number of RKNNs of a node is not known in advance. Nodes with more RKNNs are ideal pivot candidates, serving as the nearest neighbors of many others. To be specific, we first expand the neighborhood of each node $v \in V_1$ by searching from v itself for its KNNs $N_k(v)$ (nearest neighbors expansion). With $N_k(v)$ for each node v , we construct their RKNNs $R_k(v)$. Let $F_v = R_k(v) \cap U$, we define the gain of selecting v as pivots to be:

$$g(v) = \frac{|F_v|}{\sum_{u \in F_v} \delta(x_v, x_u)}$$

For simplicity, we apply a uniform search effort ef_L to all followers regardless of their distance to the pivot (see *Local Sliding* below), corresponding to treating each $\delta(x_v, x_u)$ as the maximum possible distance. The gain $g(v)$ thus simplifies to $|F_v|$, and the IDPS problem reduces to exactly the Minimum Independent Dominating Set (MIDS) problem. Since MIDS is inapproximable within $|V|^{1-\epsilon}$ for any $\epsilon > 0$ unless $P = NP$ [24, 29], we use a lazy greedy algorithm. Specifically, it maintains a max-heap ordered by $g(v)$ and greedily selects the node with the maximum gain as a pivot, assigning its uncovered RNN members as followers, until $U = \emptyset$. Since $g(v)$ can only decrease as U shrinks, the stored value in the heap is always an upper bound on the true gain. We exploit this monotonicity for efficiency via lazy evaluation: a node popped from the heap is

Algorithm 2: Reverse Neighbor Sliding Merge (RNSM)

Input: Source Graph $G_1 = (V_1, E_1)$, Target Graph $G_2 = (V_2, E_2)$, Expansion Factor k^+ , Reverse k , Naive Search ef_N , Local Sliding ef_L

Output: Merged Graph $G = (V, E)$

```

1  $V \leftarrow V_1 \cup V_2; E \leftarrow E_1 \cup E_2;$ 
2  $P \leftarrow \emptyset;$  // Set of Pivots
3  $Covered \leftarrow \emptyset;$  // Set of merged nodes
/* Phase 1: Pre-computation */
4  $G'_1 \leftarrow \text{ExpandNeighbors}(G_1, k^+);$ 
5  $\{R_k(x) \mid \forall x \in V_1\} \leftarrow \text{BuildReverseIndex}(G'_1, k);$ 
/* Phase 2: Pivot Selection (Lazy Greedy) */
6 Initialize max-heap  $H$  with  $(|R_k(x)|, x)$  for all  $x \in V_1;$ 
7 while  $H \neq \emptyset$  do
8    $(g, x) \leftarrow$  uncovered  $x$  with maximum gain;
9   if updated gain  $g' < g$  then
10      $\text{ReInsert}(g', x);$  continue;
11    $F_x \leftarrow R_k(x) \setminus Covered;$ 
12    $P \leftarrow P \cup \{x\}; Covered \leftarrow Covered \cup \{x\} \cup R_k(x);$ 
/* Phase 3: Parallel Merging */
13 for  $p \in P$  in parallel do
14   // Pivot: Naive Search
15    $Resp \leftarrow \text{SearchGraph}(G_2, \text{query} = p, \text{entry} = \text{default}, ef_N);$ 
16    $\text{UpdateGraph}(G, p, Resp);$ 
17   // Followers: Local Sliding
18   for  $y \in F_p$  do
19      $Res_y \leftarrow \text{SearchGraph}(G_2, \text{query} = y, \text{entry} = Resp, ef_L);$ 
20      $\text{UpdateGraph}(G, y, Res_y);$ 
21 return  $G;$ 

```

selected immediately if its gain is still current, and reinserted with the updated gain otherwise, avoiding redundant recomputation.

Local Sliding. Next, with each node classified into pivots and followers, we start efficient merging with local sliding. Specifically, each pivot performs a naive search from the default global entry point of G_2 , which requires a larger search effort ef_N to compensate for the potentially large distance to the target region. Once the naive search for pivot p completes, all followers in F_p proceed to local sliding using the naive search results $Resp$ as entry points. Since each follower $v \in F_p$ has pivot p as one of its KNNs in G_1 , $Resp$ in G_2 are likely to be close to the true neighbors of v , as the neighbors of a neighbor are likely neighbors [13]. This yields high-quality entry points for local sliding, allowing a much smaller ef_L to suffice. Compared to naive search for all nodes with effort ef_N , local sliding reduces the total search cost by a factor of

$$(1 - |P|/|V_1|) \cdot (1 - ef_L/ef_N)$$

Finally, we update the merged graph by applying heuristic edge selection to the search/sliding results. For each source node, its candidates are merged with its existing neighbors and pruned to retain the best R edges. The target index nodes acquire cross-partition connections solely via reverse edges inserted during this process, without performing explicit searches of their own. Empirically, this asymmetric strategy incurs negligible quality loss while halving the merge cost. The details of RNSM are presented in Algorithm 2.

- (1) **Neighbor Expansion (Line 4):** We first densify the source graph G_1 . Existing PGs are sparse approximations of the true KNN graph. We expand the neighborhood of each node x to a larger size k^+ by traversing the graph. This creates a richer set of candidate connections, ensuring that nodes can access their true nearest pivots, aligning with the IDPS objective.
- (2) **Reverse Inversion (Line 5):** We construct the RNN set $R_k(x) = \{y \mid x \in N_k(y)\}$ for every node with a relative small k (i.e., 3,5,7).

- (3) **Pivot Selection (Lines 6-12):** We maintain a max-heap ordered by marginal gain $|R_k(x) \setminus Covered|$. At each step, we extract the top candidate; if its stored gain is stale, we reinsert it with the updated gain (lazy evaluation). Once the gain is confirmed current, we select x as a pivot and assign all uncovered $R_k(x)$ to slide from x .
- (4) **Parallel Execution (Lines 11-16):** Finally, we execute the merge in parallel. Pivots perform a *Naive Search* on G_2 . Their search results become the entry points for the *Local Sliding* of their followers.

It is worth noting that both **Reverse Inversion** and **Pivot Selection** account for a negligible fraction of the total merge time (typically less than 5%), as the merge process is compute-intensive and dominated by distance computations during **Neighbor Expansion** and **Parallel Execution**. A detailed runtime breakdown of each RNSM phase is provided in the Appendix.

EXAMPLE 3. As illustrated in Fig. 7, we merge node 8 into G_2 . We initialize RNNs of each node using AKNNs obtained via nearest neighbor expansion in G_1 . Nodes 5 and 8 are selected as pivots based on their higher RNN counts, while node 4 is excluded due to coverage by pivot 5. We then perform a naive search for pivot 8: we take node 8 as the query and search from the default entry point b in G_2 via Algorithm 1, retrieving the AKNN result i . Since RNN 10 is close to node 8 in G_1 , its AKNN in G_2 should be close to i as well. Exploiting this property, i serves as an effective local entry point. We then perform local sliding from i for node 10: we search for node 10 in G_2 starting from i , and obtain its AKNN j with significantly fewer hops.

4.3 Performance Analysis

Advantages of RNSM. Using RKNNs yields several structural benefits: (1) **Natural Hub Identification:** The metric $|R_k(x)|$ identifies density peaks (hubs) purely from graph topology without requiring expensive clustering or centroid computations. (2) **Dynamic Load Balancing:** Unlike fixed k approaches, RNSM allows variable cluster sizes. Dense regions generate pivots with large $|R_k(x)|$, amortizing the naive search cost over more nodes, while sparse regions degenerate gracefully. (3) **Cost Minimization:** By selecting pivots with the most extensive coverage first, we greedily minimize the number of required pivots $|P|$, directly optimizing the naive search term in our cost model.

Complexity Analysis. We analyze the time complexity of merging a source graph index G_1 into a target graph index G_2 to produce a unified index $G = (V, E)$, where $n = |V|$ denotes the total number of vertices. Without loss of generality, we assume the source index is not larger than the target index (i.e., $|G_1| \leq |G_2|$). If initially $|G_1| > |G_2|$, the two are swapped so that we always merge the smaller structure into the larger one—a common optimization that speeds up merging. A key property of these graph-based indices, established in prior work [16, 28], is that the maximum node degree is bounded by a constant. This sparsity guarantee is crucial for achieving an efficient merge procedure. The merging process consists of three sequential stages:

1. **Neighbor Expansion:** Expanding neighbors involves incremental search with the graph index. Start with the nearest neighbor;

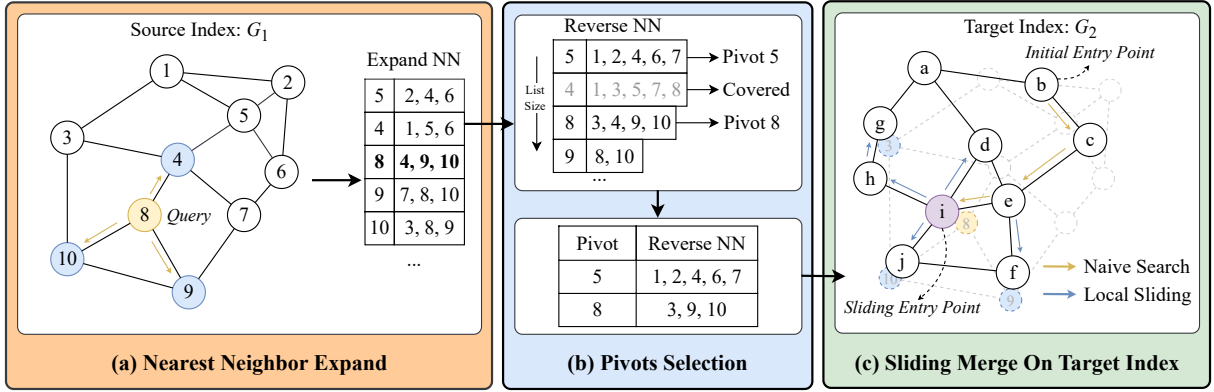


Figure 7: Example of Two-Index Merging with RNSM

the incremental cost of searching the top- k' neighbors can be expressed as $O(k')$ with a small constant factor [69]. This efficiency stems from interpreting the existing proximity graph (PG) index as a compressed representation of the k -nearest neighbor graph [16]. *Total Complexity:* $O(n \cdot k^+)$.

2. Pivot Selection: Constructing the RKNN graph requires iterating over all edges in the expanded graph. Given that the total number of edges is $n \cdot k^+$, the inversion step takes $O(n \cdot k^+)$ time. Subsequently, sorting the nodes on the RNN graph using bucket sort can be performed in $O(|V| + |E|)$ time. Since $|V| = n$ and $|E| = n \cdot k < n \cdot k^+$, the *total complexity* of the entire process remains $O(n \cdot k^+)$.

3. Sliding Merge: Let $|P|$ be the number of selected pivots.

- **Pivots:** Perform naive AKNN search. In typical graph structures (NSG, HNSW), the search complexity is bounded by $c_{naive} = O(\log n \cdot ef \cdot d)$, where ef is the search parameter and d is the average degree. Total: $O(|P| \cdot c_{naive})$.
- **Followers:** Perform local sliding. Since the pivot $p \in N_k(y)$ by construction, the sliding path length is proportional to the distance $\delta(p, y)$ and is small. In practice, the search complexity is bounded by a small constant c_{slide} . Total: $O((n - |P|) \cdot c_{slide})$.

Overall Complexity: Both local sliding and naive search have a higher cost than neighbor expand, where $c_{naive} > c_{slide} > k^+$. The overall complexity can be derived to $O(|P|c_{naive} + (n - |P|)c_{slide})$. Since $|P| < n$ (depending on k , typically 5% to 10% of nodes), the dominant term is reduced from $n \cdot c_{naive}$ (all-naive search) to roughly $n \cdot c_{slide}$, representing a significant speedup.

5 MOS for Multiple Indexes Merge

In this section, we minimize the multi-index merge cost by reducing redundant operations. We first formalize the merge order selection task. Then, we propose algorithms for our formalized problem.

Merge Order Selection. Recall that merging requires finding the KNNs in all other indexes for every node in each source index. Naive approach requires $O(m^2)$ index merge operations between m indexes. However, this approach contains significant redundancy in practice. Besides, our empirical analysis (Fig. 4) shows that different merge order strategies yield substantially different search performance. Thus, we formally define the task as follows: Given m

dataset partitions $\{P_1, P_2, \dots, P_m\}$, we represent partitions as vertices $V = \{v_1, \dots, v_m\}$ and merge operations as edges E , forming a merge order graph $\mathbb{G} = (V, E)$. Each edge $(v_i, v_j) \in E$ indicates that we perform index merging between index G_i and G_j . The merge order selection (MOS) task aims to adjust the merge order graph $\mathbb{G}$ to achieve better individual sub-index connections at low cost.

Cost Model. Given the notation above, our objective is to extract a subgraph from the complete graph of pairwise merges. We aim to identify a merge schedule—defined by this subgraph—that achieves search performance comparable to either a full index rebuild or an exhaustive pairwise merge strategy. For clarity, we analyze random and clustered partitions separately. For random partitions, we assume that merging any two partitions incurs a unit cost. For clustered partitions, the merge cost between two partitions depends on the distance between their centroids, as derived in § 4.1. Moreover, § 3.1 indicates that Path, Star, and MST-based merge strategies can lead to substantially different search performance. These differences can be understood through graph-theoretic design properties such as connectivity, diameter, and total edge weight. Unfortunately, relying on a single subgraph perspective did not achieve our expected superior search performance.

MOS Problem. Therefore, we propose extracting subgraphs from multiple perspectives with minimal cost, specifically:

Definition 5.1 (Minimum Cost Merge Order Selection (MOS)).

- **Input:** Given a set of m vertices $V = \{v_1, v_2, \dots, v_m\}$ and a symmetric cost matrix C where C_{ij} is the cost of connecting vertex v_i and v_j , diameter bound Δ and degree bound R .
- **Output:** An undirected graph $\mathbb{G} = (V, E)$ such that:
- **Constraint:** (1) **Connectivity:** The graph $\mathbb{G}$ is connected. (2) **Maximum Degree:** For all vertices $v \in V$, the degree $\deg(v) \leq R$. (3) **Graph Diameter:** The diameter of $\mathbb{G}$ is at most Δ .
- **Cost Objective:** Minimize the total cost of the graph:

$$\min \sum_{(v_i, v_j) \in E} C_{ij}$$

The constraints serve distinct objectives:

- (1) **Connectivity:** This is the fundamental completeness requirement, ensuring that the search space is fully traversable.
- (2) **Degree Bound (R):** This constraint parallel-friendly design. The index merge operation locks a large proportion proximity

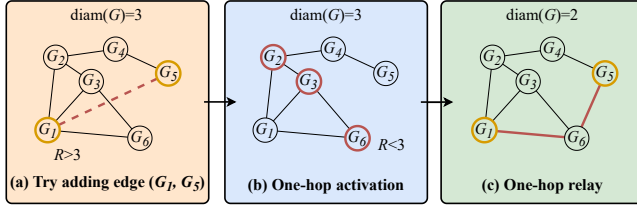


Figure 8: Example of Merge Order Selection

graph node neighbors. A vertex with excessive degree serializes updates and causes the parallel workload imbalance.

- (3) **Diameter Bound (Δ):** This constraint governs the worst-case search latency. The diameter determines the maximum number of intermediate index hops required to traverse between any two partitions.

While minimizing the total edge weight reduces the direct merge costs, the graph edge weights also affect the runtime performance from our empirical analysis in § 4.3. Ignoring the diameter bound reduces the problem to the NP-hard Bounded Degree Minimum Spanning Tree Problem [33]. To address this NP-hard problem efficiently, we propose a greedy heuristic as follows. We initialize the merge order graph as an empty structure and iteratively insert nodes to satisfy the Connectivity constraint. Subsequently, for each node, we establish connections with nodes located beyond Δ -hop distance (typically $\Delta=2$) to ensure compliance with the Diameter Bound. When either node possesses R neighbors, we relay the connection to its $(\Delta - 1)$ -hop neighbors to meet both Degree and Diameter Bound. Throughout this process, connections are prioritized according to the cost matrix, with lower-cost pairs being added first. The detailed MOS algorithm is presented below.

MOS Algorithm. Algorithm 3 introduces the detailed procedure of MOS. We begin with an empty graph $\mathbb{G}$ and iteratively add edges (lines 1-2). We process vertices $v \in V$ in ascending order of their average distance to the R nearest centroids, prioritizing centrally-located partitions (lines 3-5). For each v , we iteratively connect it to its nearest neighbor x that is more than Δ hops away (line 7). If both v and x have degree less than R , we directly add edge (v, x) (lines 8-9). If exactly one vertex has reached degree limit R , we employ a 1-hop relay strategy (lines 10-14), select a neighbor x' of the degree-saturated vertex, and connect it to the unsaturated one. Note that the subgraph may not exist due to the Moore bound [26]; however, we still enforce connectivity in practice for result completeness.

EXAMPLE 4. As illustrated in Fig. 8, MOS constructs a workload-balanced merge order graph with diameter 2 by inserting and relaying edges within 1-hop neighborhoods. Suppose we insert edge (G_1, G_5) when the degree limit of 3 is exceeded for G_1 (i.e., $\deg(G_1) = 4$). We then relay this connection to one of the 1-hop neighbors of G_1 , and G_6 is selected since $\deg(G_6) < 3$. By relaying edge (G_1, G_5) as (G_5, G_6) , we obtain a graph with maximum degree at most 3 and diameter 2.

Complexity. We now analyze the time complexity of Algorithm 3. The ordering phase requires $O(m^2 \log R)$ for computing average distances of R nearest neighbors and sorting vertices. The main loop adds $O(m \cdot R)$ edges total, with each edge addition requiring $O(m \cdot R)$ time for BFS-based incremental hop-distance matrix updates,

Algorithm 3: Merge Order Selection MOS

Input: Partition Centroids $\{c_1, c_2, \dots, c_m\}$, Cost Matrix C , Degree Bound R , Diameter Bound $\Delta = 2$
Output: Merge order graph $\mathbb{G} = (V, E)$

```

1  $V \leftarrow \{c_1, c_2, \dots, c_m\}; E \leftarrow \emptyset;$ 
2  $H \leftarrow$  distance matrix in hops, initialized to  $\infty$ ;  $Distances \leftarrow \emptyset;$ 
3 for  $c_i \in V$  do
4    $Distances[c_i] \leftarrow$  average distance of  $R$  nearest centroids;
5 order  $\leftarrow V$  sorted in ascending order of  $Distances$ ;
6 for  $v \in \text{order}$  do
7   while has nearest vertex  $x$  with  $H[v][x] > \Delta$  do
8     if  $\deg(v) < R$  and  $\deg(x) < R$  then
9        $E \leftarrow E \cup (v, x);$ 
10    else if  $\deg(v) < R$  and  $\deg(x) \geq R$  then
11       $x' \leftarrow \arg \min_{x' \in \text{Neighbor}(x), \deg(x') < R} C_{x', v}; E \leftarrow E \cup (v, x');$ 
12    else if  $\deg(v) \geq R$  and  $\deg(x) < R$  then
13       $v' \leftarrow \arg \min_{v' \in \text{Neighbor}(v), \deg(v') < R} C_{v', x}; E \leftarrow E \cup (v', x);$ 
14    Update  $H$ ;
15 return  $G$ ;
```

yielding a complexity of $O(m^2 R^2)$. Overall, the time complexity of our MOS algorithm is $O(m^2 R^2)$.

Note that m and R are typically much smaller than the dataset size n and the dimensionality d , so distance computation remains the performance bottleneck. Despite the high complexity of updating the hop distance matrix, in practice, the number of partitions and the degree bound are relatively small. Even with around 100 partitions, the overall time cost of the MOS algorithm is only 1-2 seconds, which is negligible compared to the index construction time.

6 Experiment

6.1 Experiment Setting.

Datasets. We use six public datasets with diverse scales and types. Including datasets that are widely used in the benchmarks (SIFT, DEEP, GIST)¹ and datasets generated from the latest embedding models (IMAGE², ANTON³, MARCO⁴). The statistics of those datasets are summarized in Table 2, where *Base Size* denotes the dataset cardinality and *Query Size* denotes the number of queries for testing. We also take several slices of distinct sizes from the original datasets to test the scalability of our framework. For example, we use DEEP1M to represent 1 million slices of the DEEP dataset, and use ANTON10M to indicate the 10 million scale in the experiment.

Methods. We compare the following index merge methods:

- **Rebuild:** Rebuild the complete proximity graph directly by incrementally inserting nodes into an empty graph.
- **CM:** Cross-query Merge (CM) using naive search. It uses a smaller candidate set $ef \ll ef_{\text{construction}}$ and one-way search (only target index searches source index, not bidirectional), which are the two key optimizations used in HNSW-merger [32].
- **FGIM:** Method for merging PG indexes and transforming it into a KNN graphs merging and refining problem [64].
- **Overlap:** Overlapping-based method that merges indexes through duplicate data points [52].

¹<https://www.cse.cuhk.edu.hk/systems/hash/gqr/datasets.html>

²https://huggingface.co/datasets/kinianlo/imagenet_embeddings

³<https://huggingface.co/datasets/anton-l/wiki-embed-mxbai-embed-large-v1>

⁴<https://huggingface.co/datasets/CoHere/msmarco-v2.1-embed-english-v3>

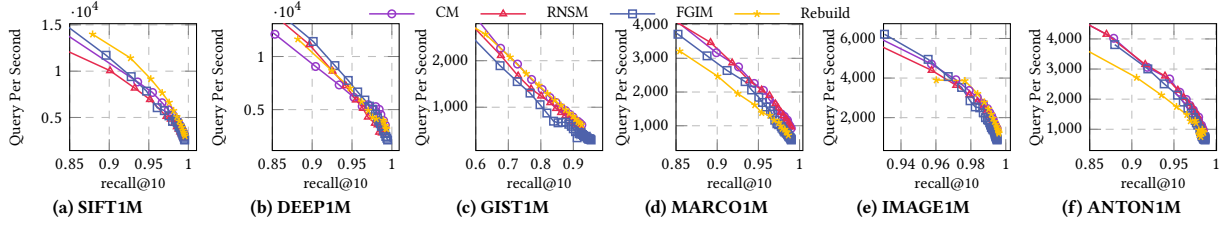


Figure 9: Recall@10-Query Per Second Tradeoff of Various Method

Table 2: Dataset Statistics

Dataset	Dimension	Base Size	Query Size	LID [6]
SIFT	128	1,000,000	10,000	19.4
DEEP	96	100,000,000	1,000	19.5
GIST	960	1000,000	1,000	38.9
IMAGE	768	10,000,000	10,000	21.2
ANTON	1024	10,000,000	1,000	28.9
MARCO	1024	10,000,000	1,000	31.0

• **RNSM**: Our Reverse Neighbor Sliding Merge (RNSM) method without Merge Order Selection (MOS).

• **RNSM***: Our RNSM with MOS for random partitions.

• **RNSM⁺**: Our RNSM with MOS for cluster partitions.

Performance Metrics. We evaluate the merged graph index based on the following two primary metrics:

- **Search Performance:** We measure efficiency via Queries Per Second (QPS) and accuracy via Recall@K as defined in § 2. Consistent with standard practice, we report the average Recall@10 and QPS with the top 10 AKNN search over the query set.
- **Merge Efficiency:** We quantify efficiency using the speed-up ratio relative to the CM, FGIM, *Rebuild* baselines, and the overlapping construction strategy employed by DiskANN.

Parameter Setting. We integrate our methods with various graph-based indexing algorithms, each with its own parameters. We leave the other parameters as default [60] and only consider the beam search candidate set size ef . In particular, we examine several widely adopted graph-based AKNN search algorithms: HNSW [39], NSG [16], SSG [15], and r -MNG [44]. To determine the parameters ef_N in Algorithm 2, we set it as the default candidate size that achieves an average Recall@10 > 0.98 on separated indexes based on observation of Fig. 3. Then, we employ grid search to determine the optimal parameter values of k^+ , k , and ef_L for Algorithm 2. For fair comparison, we set the same beam search parameter ef and max PG node degree R used for CM, and RNSM. The parameter of FGIM is set according to its default settings [64], where $ef = \frac{R}{m-1}$.

Implementation Environment. All experiments were conducted on a server running Ubuntu 22.04 LTS, equipped with dual Intel Xeon Platinum 8370C CPUs (2.80GHz, 32 physical cores each, supporting AVX-512) and 512 GB of DRAM. The proposed framework and baselines were implemented in C++ based on the hnsplib library [40] and compiled using GCC 13.1.0 with -O3 optimization.

6.2 Experiments Results

Exp-1: Test of Indexes Merging Efficiency. We begin by evaluating the merging efficiency and subsequent search performance of our RNSM framework. For the setup, we randomly partition each dataset in Table 2 into two disjoint subsets of 0.5 million vectors

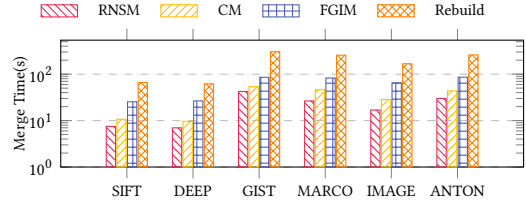


Figure 10: Index Merge Efficiency Speedup

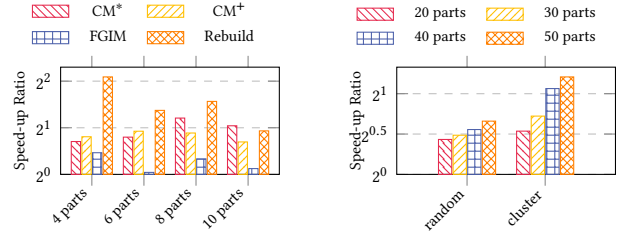


Figure 11: Speed-up Ratio Comparison across Different Partitions with MARCO10M

and construct indexes using hnsplib. We compare RNSM against CM, FGIM, and a reconstruct-from-scratch baseline *Rebuild*.

Fig. 10 reports the speed-up ratio of RNSM over these methods across six datasets. RNSM achieves significant efficiency gains over all baselines. For example, RNSM delivers speed-up ratios up to 1.74× over CM, 3.86× over FGIM, and 9.92× over *Rebuild*. Notably, CM even outperforms the FGIM. According to Theorem 4.1 in FGIM [64], when the number of partitions is 2, the cross-index search in FGIM reduces to a naive search of CM. However, FGIM incurs additional overhead from refining KNN graphs and transforming the KNN graph into PG indexes.

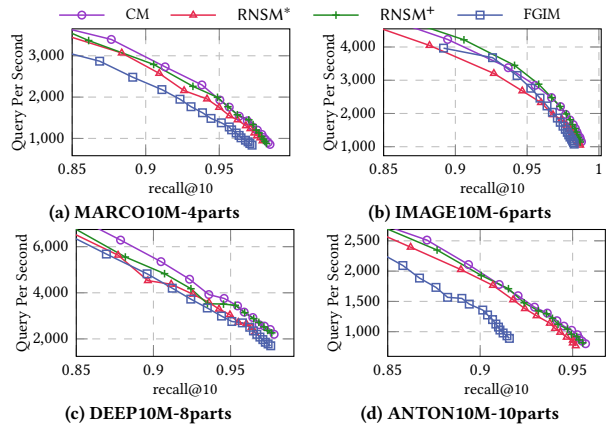


Figure 13: Recall@10-QPS Tradeoff for Large Scale Dataset

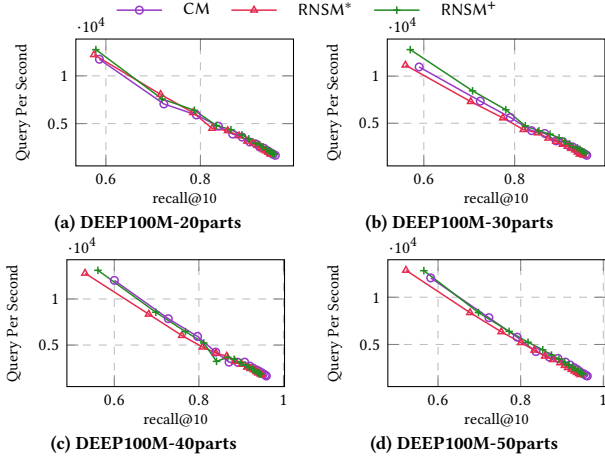


Figure 14: Recall@10-QPS Tradeoff for More Data Partitions

Fig. 9 further demonstrates the search performance (Recall vs. QPS). RNSM achieves a trade-off comparable to existing methods and the *Rebuild* baseline on all datasets. These results confirm that our framework substantially improves merge efficiency without compromising the search performance of the resulting index.

Exp-2: Test of Multiple Indexes Merging. We further evaluate the search performance and merge efficiency when merging multiple indexes. We vary the number of partitions ($\{4, 6, 8, 10\}$) and assess robustness across different data distributions using both random (RNSM*) and cluster (RNSM+) partitioning. Experiments are conducted on DEEP10M, MARCO10M, IMAGE10M, and ANTON10M. Note that we compare RNSM* against both CM and FGIM, but compare RNSM+ only against CM, as FGIM is incompatible with cluster partitions. Fig. 11 reports the merge efficiency on MARCO10M (full results in Appendix). Our methods achieve substantial speedups over baselines. For example, on MARCO10M, RNSM* delivers speed-up ratios of up to $2.31\times$ over CM, $1.38\times$ over FGIM, and $4.26\times$ over *Rebuild*, while RNSM+ achieves a $1.90\times$ speed-up over CM. Regarding search quality, as shown in Fig. 13, our framework maintains performance comparable to baselines across varying partition numbers and distributions.

Exp-3: Comparison with Overlap-based Method. We compare our framework against the overlapping partition strategy used in DiskANN on 10M-scale datasets. Unlike DiskANN, which partitions data into overlapping shards (exactly one duplication for each vector) and builds sub-indexes from scratch, RNSM directly merges existing sub-indexes. For a fair comparison, we align the maximum shard size of Overlap with the total vector count of the indexes merged by RNSM (e.g., if Overlap uses a 5M shard size, RNSM merges four 2.5M sub-indexes). We also use the same index of Overlap strategy to focus on the merge algorithm itself. We include both partitioning and construction time for both methods and report the QPS with 0.95 recall in Table 3. Our method achieves comparable search efficiency while significantly reducing index construction time. Specifically, RNSM is consistently faster across all datasets even without any pre-built indexes, delivering speedups of up to $2.01\times$ compared to the scratch-build approach

Table 3: Comparison of DiskANN and RNSM+. (Time: seconds, QPS_x: Query Per Second with recall@10=x)

Max Size	Dataset	Overlap DiskANN			RNSM*		
		Time	QPS _{0.90}	QPS _{0.95}	Time	QPS _{0.90}	QPS _{0.95}
5M	DEEP10M	1247	4538	3042	630	4554	3175
	ANTON10M	4116	1672	783	2928	1637	819
	MARCO10M	4410	2021	1436	3138	2573	1751
	IMAGE10M	2575	3509	2314	1518	3423	2395
2.5M	DEEP10M	1206	5295	3225	767	4901	3493
	ANTON10M	4084	1803	780	2649	1791	797
	MARCO10M	4446	2308	1528	3295	2429	1662
	IMAGE10M	3185	3603	2550	1582	3807	2456

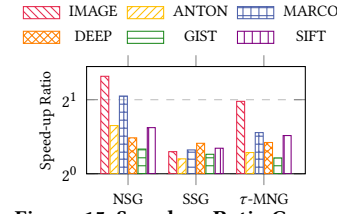


Figure 15: Speed-up Ratio Comparison of Different Methods

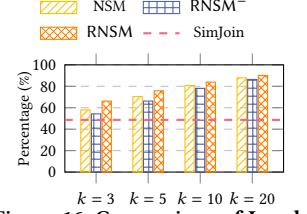


Figure 16: Comparison of Local Sliding Ratio with DEEP10M

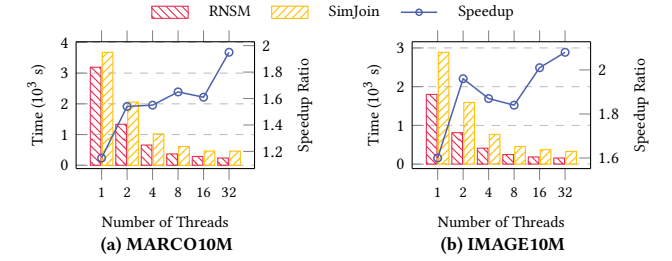


Figure 17: Parallel Scalability Comparison

of DiskANN. Furthermore, the index construction quality is better than the Overlap method, resulting in a higher QPS at 0.95 recall.

Exp-4: Scalability Study for Index Merging. We assess the scalability of our methods regarding dataset size and partition count on the large-scale DEEP100M dataset. We partition the data into $\{20, 30, 40, 50\}$ subsets using both random and cluster partitioning and merge them using RNSM* and RNSM+, respectively. We exclude FGIM here as its theoretical guarantees do not hold when the partition number approaches the maximum degree (usually 32) of the sub-indexes. Results are shown in Fig. 12 and Fig. 14. RNSM consistently outperforms the CM baseline without compromising search performance. Notably, the efficiency advantage of our framework amplifies as the system scales: the speedup ratio grows as the number of partitions increases, demonstrating that RNSM is well-suited for massive multi-index merging scenarios.

Exp-5: Applicability Study. To demonstrate the generality of RNSM, we evaluate on three additional graph-based index structures: NSG, SSG, and τ -MNG. We configure NSG and SSG following recommendations in [60] and implement τ -MNG on the NSG codebase [44]. Fig. 15 presents the merge efficiency. RNSM achieves consistent speedups across all index types. For example, it delivers speedups of up to $2.49\times$ for NSG, $1.33\times$ for SSG, and $1.97\times$ for τ -MNG. Furthermore, our results confirm that RNSM maintains

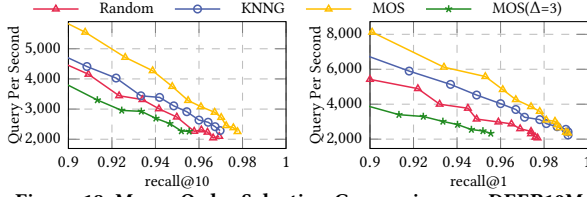


Figure 18: Merge Order Selection Comparison on DEEP10M

search quality comparable to CM across these diverse structures (see Appendix), validating the broad applicability of RNSM.

Exp-6: Effects of Reverse Neighbor Pivots Selection. We analyze the impact of using RKNNs for pivot selection. By selecting pivots with more RKNNs, we maximize the number of nodes processed via local sliding rather than naive search. We compare four variants: (1) NSM: randomly select pivots, slide to KNNs; (2) SimJoin: MST roots as pivots, slide to internal nodes and leaves; (3) RNSM⁻: randomly select pivots, slide to RKNNs; and (4) RNSM: pivots and corresponding followers selected by our greedy algorithm. We evaluate these strategies on SIFT1M with $k \in \{3, 5, 10, 20\}$ and report the proportion of local sliding operations in Fig. 16. Notably, SimJoin exhibits the lowest local sliding ratio among all compared methods. This limitation arises because its ratio is strictly determined by the underlying graph index and the dataset, making it incapable of dynamically adjusting the sliding proportion. For the other three methods, the local sliding ratio grows significantly as k increases. For RNSM, the ratio rises from 66.2% at $k=3$ to 90.1% at $k=20$, confirming that higher connectivity naturally facilitates more local moves. RNSM consistently outperforms both NSM and RNSM⁻ across all k values. This demonstrates that prioritizing hubs (nodes with higher reverse nearest neighbor cardinality) as pivots allows the algorithm to cover more points through efficient sliding, thereby minimizing expensive naive searches. RNSM⁻ performs worse than NSM, since NSM iteratively propagates search results from the last iteration as local entry points instead of sliding to a fixed number of nodes. However, by simply introducing the sorting strategy, RNSM surpasses NSM. This indicates that while RKNN-based sliding is a powerful tool, its success is highly dependent on the pivot selection.

Exp-7: Parallel Scalability. We further compare RNSM and SimJoin under varying thread numbers to evaluate the parallel efficiency of our pivot-based sliding framework. Experiments are conducted on MARCO10M and IMAGE10M with $\{1, 2, 4, 8, 16, 32\}$ threads, with both methods configured to achieve comparable merged index search quality. As shown in Fig. 17, RNSM consistently outperforms SimJoin across all thread configurations, with its relative advantage increasing with parallelism. On MARCO10M, the speedup improves from 1.15 $\times$ (single-threaded) to 1.95 $\times$ (32 threads), and from 1.60 $\times$ to 2.08 $\times$ on IMAGE10M. This discrepancy is attributable to a structural distinction between the two methods. SimJoin propagates search results along a pre-built MST, introducing strict inter-node dependencies that limit parallelism. Conversely, RNSM decouples local sliding operations via pivot selection, thereby enabling fully independent processing.

Exp-8: Effects of MOS Technique. Finally, we validate the effectiveness of the MOS strategy for merging multiple partitions. We compare the search performance of our Merge Order Graph

against a random regular graph and a KNNG with the same average degree. We also compared with a MOS graph that alternates the diameter bound to $\Delta = 3$. Using simple merge orders with fewer merge operations, as employed in our motivations like MST, Star, and Path, fails to achieve a search recall of 0.9 and is therefore not considered in our experiment. Experiments are conducted on DEEP10M split into 10 partitions. As shown in Fig. 18, the topology generated by MOS consistently yields better search performance than the random graph.

7 Related Works

HNSW-merger. HNSW-merger [32] proposed merging HNSW indexes via cross-querying. Specifically, they demonstrate that the merged index achieves search performance comparable to a full rebuild even with a small candidate set ef and one-way search (only search the larger index to find neighbors for nodes of the smaller index), which is consistent with our Observation 1. To further reduce overhead, they utilize the special hierarchical structure of HNSW to merge layer by layer and cache traversal entry points from higher layers. However, they handle multi-index merges in the manner of incremental insert by repeatedly merging the two largest indexes, limiting speedups.

Similarity Join. SimJoin [65] was originally proposed to efficiently compute similarity joins, outputting all pairs of vectors from two datasets whose distance falls below a specified threshold. This task is intrinsically linked to index merging: identifying cross-index KNNs between two partitions is effectively a similarity join operation restricted to the top- k nearest neighbors. However, it requires constructing an MST with one of the partitions and cannot be directly applied to merging two PG indexes.

KNNG Merge. S-merge [76] proposes merging two k -Nearest Neighbor Graphs (KNNG) by first connecting them with random edges and then iteratively refining the graph using NNDecent [13]. Multi-way Merge [73] extends this concept to merge multiple KNNGs in a single pass by cross-matching only newly added cross-index candidates. Generally, the KNNG quality converges faster than PG-based methods due to the batch processing of NNDecent [72]. However, KNNGs are not optimized for AKNN search and typically yield inferior search performance compared to navigational graph indexes [4, 36, 60, 61].

8 Conclusion

In this paper, we addressed the challenge of merging multiple proximity graph indexes to enable efficient multi-index searching. By identifying the core components of graph merging, we proposed the Reverse Neighbor Sliding Merge (RNSM) algorithm to facilitate pairwise merge. Furthermore, based on merge order analysis, we designed the Merge Order Selection (MOS) algorithm. MOS is suitable for both random and cluster partitions, and it reduces redundancy in pairwise processing by pruning unnecessary merges. Our experimental results demonstrate that our framework significantly improves merging efficiency while preserving the superior search performance characteristic of graph indexes. Notably, our method exhibits robust scalability, performing effectively even with large-scale datasets.

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Appendix

Proof

Proof of Theorem 4.2

PROOF. To prove that the Minimum Cost Independent Dominating Pivot Selection (IDPS) problem is NP-hard, we establish a polynomial-time reduction from the NP-complete **Minimum Independent Dominating Set (MIDS)** problem [29].

First, we define the decision versions of both problems.

- **MIDS-Decision:** Given an undirected graph $G' = (V', E')$ and a positive integer k , does there exist an independent dominating set $D \subseteq V'$ with $|D| \leq k$? (i.e., every $v \notin D$ has a neighbor in D , and no two nodes in D are adjacent.)
- **IDPS-Decision:** Given a vector dataset X , a graph $G = (V, E)$, a distance function $\delta(\cdot, \cdot)$, a naive search cost γ , and a target cost C , does there exist a pivot subset $P \subseteq V$ satisfying the independent dominating set constraint such that $\mathcal{L}(P) \leq C$?

NP Membership. IDPS-Decision is in NP: given a candidate P , one can verify in polynomial time that (i) every $v \notin P$ has a neighbor in P (domination), (ii) no two nodes in P are adjacent (independence), and (iii) $\mathcal{L}(P) \leq C$.

Polynomial-Time Construction. Given an instance $\langle G', k \rangle$ of MIDS-Decision, we construct an instance $\langle X, G, \delta, \gamma, C \rangle$ of IDPS-Decision as follows:

- (1) Let $G = G'$, so $V = V'$ and $E = E'$.
- (2) Map each node $v_i \in V$ to a distinct vector $x_i \in X$. Define the discrete metric: $\delta(x_u, x_u) = 0$ and $\delta(x_u, x_v) = 1$ for all $u \neq v$. (This trivially satisfies all metric space properties.)
- (3) Set $\gamma = 2$ and $C = k + |V|$.

This construction runs in polynomial time $\mathcal{O}(|V|^2)$.

Cost Simplification. For any P satisfying the independent dominating set constraint, P is in particular a dominating set, so every $v \notin P$ has a neighbor $p \in P \cap N(v, G)$. With $\delta \equiv 1$ for distinct nodes, the sliding cost is:

$$\mathcal{L}_{slide}(P) = \sum_{v \in V \setminus P} \min_{p \in P \cap N(v, G)} \delta(x_v, x_p) = |V| - |P|$$

Substituting $\gamma = 2$:

$$\mathcal{L}(P) = 2|P| + (|V| - |P|) = |P| + |V|$$

Correctness of Reduction.

- $(\Rightarrow)$ Suppose there exists an independent dominating set D in G' with $|D| \leq k$. Let $P = D$. Since $G = G'$, P is an independent dominating set in G , satisfying both IDPS constraints. The total cost is $\mathcal{L}(P) = |P| + |V| \leq k + |V| = C$. Thus, a valid IDPS solution exists.
- $(\Leftarrow)$ Suppose there exists a pivot subset $P \subseteq V$ satisfying the IDPS constraints with $\mathcal{L}(P) \leq C$. By the IDPS constraints, P is an independent dominating set of $G = G'$. From $\mathcal{L}(P) \leq C$ we get $|P| + |V| \leq k + |V|$, so $|P| \leq k$. Thus, an independent dominating set of size at most k exists in G' .

Since MIDS-Decision is NP-complete and MIDS-Decision $\leq_p$ IDPS-Decision, IDPS-Decision is NP-complete. Consequently, the IDPS optimization problem is NP-hard. $\square$

Extra Experiments

Runtime Breakdown of RNSM In this section, we analyze the execution time of RNSM by decomposing it into its core components (32 threads): pivot selection, neighbor expansion, and merging. As shown in Fig. 19, the merging step dominates runtime across all datasets, accounting for more than 83.1% of total time, since it involves the main graph-update operations driven by the sliding cost model. Neighbor expansion contributes 8.7%–15.3%, reflecting the overhead of retrieving nearest neighbors of every node from the source index. Pivot selection is consistently negligible ($\leq 5.0\%$), confirming that the sequential pivot in Algorithm 2 adds virtually no overhead.

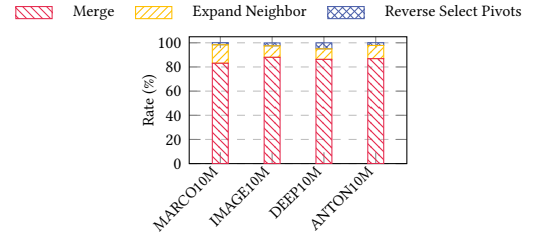


Figure 19: Time Breakdown of RNSM Components

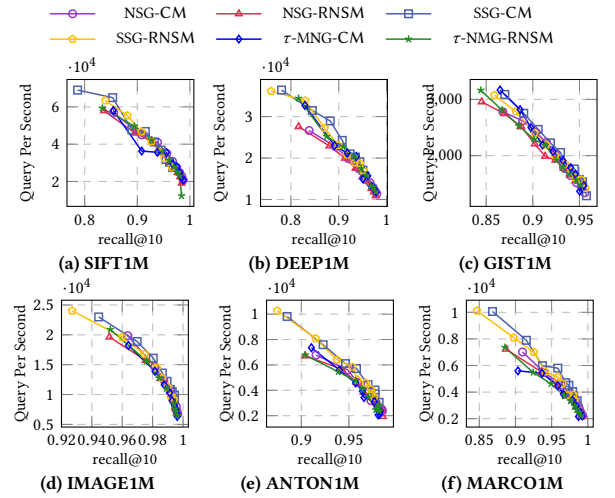


Figure 20: Recall@10-QPS Tradeoff of Different Methods

Applicability Study of Search Performance This section evaluates the applicability of RNSM across different graph index structures beyond HNSW. We compare RNSM against the Cross-query Merge (CM) baseline on three representative proximity graph types: NSG [16], SSG [15], and τ -MNG [44].

Fig. 20 presents the complete results. Across all configurations, RNSM consistently maintains search performance comparable to or better than CM. This demonstrates that RNSM preserves the navigability and search efficiency of the underlying graph indexes, regardless of their specific construction algorithms or edge selection policies.

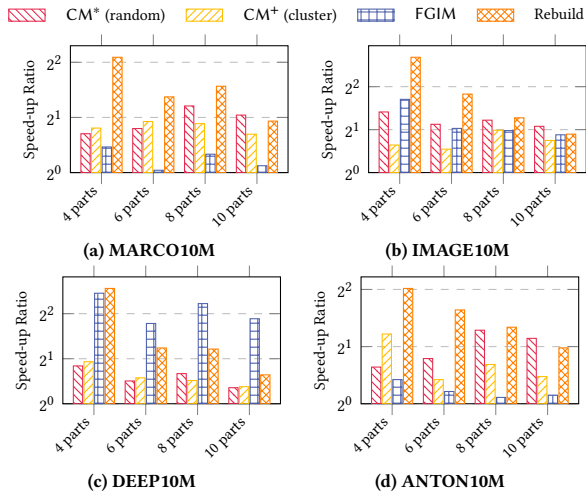


Figure 21: Speed-up Ratio Comparison across Different Partitions

Comprehensive Results for Exp-2 This section presents the comprehensive experimental results for Exp-2 (Multiple Index Merging). It covers all evaluated datasets (DEEP10M, MARCO10M, IMAGE10M, and ANTON10M) using both random and cluster partitioning strategies across varying partition counts {4, 6, 8, 10}.

Fig. 21 reports the complete speedup ratios across all datasets and partition numbers. Our method shows the best merge efficiency in all cases, achieving up to 2.33 $\times$, 5.48 $\times$, and 6.43 $\times$ acceleration compared to CM, FGIM, and BuildAsOne, respectively.

Fig. 22 presents the complete recall@10-QPS tradeoff curves for all experimental configurations. The results demonstrate that our RNSM and MOS framework maintains the expected search performance of graph indexes and consistently outperforms the existing merge method, FGIM.

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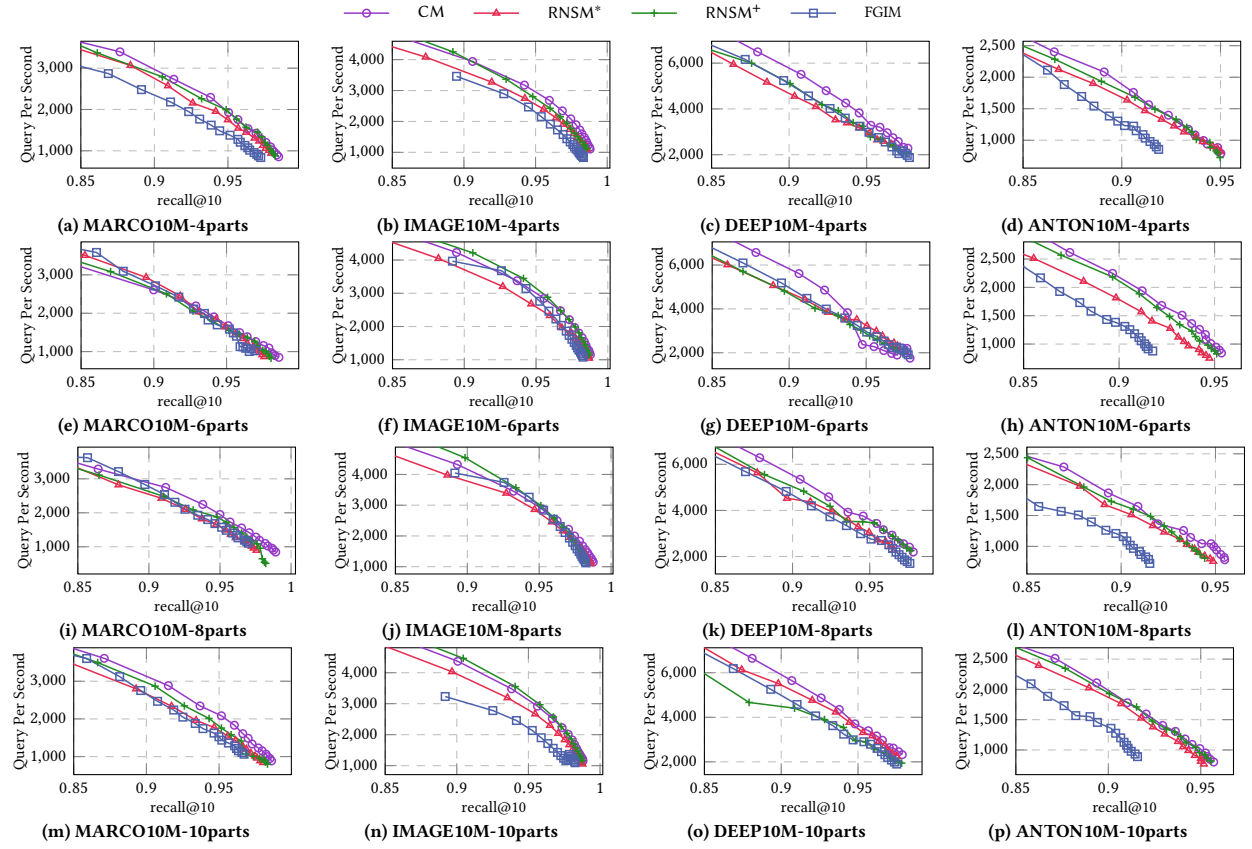


Figure 22: Recall@10-Query Per Second Tradeoff for Merge Methods